Laboratory Report 3–5

Student: ______________________

Desk No.: __________ Section: __________

Ex. 3 Phase-Contrast Microscopy

A. Questions

1. Which rays (*direct* or *diffracted*) are altered by the phase ring on the phase plate? _____
2. How much phase shift occurs in the light rays that emerge from a transparent object? _____
3. Differentiate:

   Bright phase microscope: ______________________

   Dark phase microscope: ______________________
4. List two items that can be used for observing the concentricity of the annulus and phase ring:

   a. ______________________ b. ______________________

Ex. 4 Fluorescence Microscopy

A. Questions

1. Differentiate:

   Phosphorescence: ______________________

   Fluorescence: ______________________
2. List three fluorochromes that are used in staining bacteria.

   a. ______________ b. ______________ c. ______________
3. What are the two most serious hazards when using mercury vapor arc lamps?

   a. ______________________ b. ______________________
4. What relationship exists between the wavelength of light and its energy?

   ____

Ex. 5 Microscopic Measurements

A. Questions

1. What is the distance between each of the graduations on the stage micrometer? ______________ mm
2. Why must the entire calibration procedure be performed for each objective? ______________

   ____

(See reverse page for more questions pertaining to Exercises 3 and 4.)

Phase-Contrast and Fluroscence Microscopy

B. *Multiple Choice Questions for Exercises 3 and 4*

Select the answer that best completes the following statements.

1. If direct rays passing through an object are advanced ¼ wavelength by the phase ring, the diffracted rays are
   1. in phase with the direct rays.
   2. ½ wavelength out of phase with the direct rays.
   3. ¾ wavelength out of phase with direct rays.
   4. in reverse phase with the direct rays.
   5. Both 2 and 4 are correct.
2. The best barrier filter to use with the Schott BG12 exciter filter is the
   1. blue A0702.
   2. Schott OG1.
   3. Wratten G-2.
   4. None of these are correct.
3. The examination of ordinary stained slides is not enhanced with a
   1. brightfield microscope.
   2. phase-contrast microscope.
   3. fluorescence microscope.
   4. Both 2 and 3 are correct.
4. Special immersion oil is required for
   1. brightfield microscopy.
   2. phase-contrast microscopy.
   3. fluorescence microscopy.
   4. Both 2 and 3 are correct.
5. Amplitude summation occurs in phase-contrast optics when both direct and diffracted rays are
   1. in phase.
   2. in reverse phase.
   3. off ¼ wavelength.
   4. None of these are correct.
6. The phase-contrast microscope is best suited for observing
   1. living organisms in an uncovered drop on a slide.
   2. stained slides with cover glasses.
   3. living organisms in hanging drop slide preparations.
   4. living organisms on a slide with a cover glass.
7. The wavelength of fluorescent light rays is
   1. always longer than the exciting wavelength.
   2. always shorter than the exciting wavelength.
   3. about the same length as the exciting wavelength.
   4. sometimes shorter and at other times longer in wavelength.
8. The barrier filter in the fluorescence microscope is kept in position to
   1. block all light rays from getting through.
   2. allow fluorescence light rays to pass through.
   3. screen out exciting light rays.
   4. Both 2 and 3 are correct.
9. A phase centering telescope is used to
   1. improve the resolution of the ocular.
   2. increase magnification with the oil immersion objective.
   3. observe the relationship of the annular diaphragm to the phase ring.
   4. None of the above are correct.
10. The visible spectrum of light is between
    1. 200 and 800 nanometers.
    2. 400 and 780 nanometers.
    3. 300 and 800 nanometers.
    4. None of these are correct.

Answers
Multiple Choice
1. ________
2. ________
3. ________
4. ________
5. ________
6. ________
7. ________
8. ________
9. ________
10. ________

LABORATORY REPORT 6

Student: ____________________

Desk No.: ________ Section: __________

Protozoa, Algae, and Cyanobacteria

A. *Tabulation of Observations*

In this study of freshwater microorganisms, record your observations in the following tables. The number of organisms to be identified will depend on the availability of time and materials. Your instructor will indicate the number of each type that should be recorded.

Record the genus of each identifiable type. Also, indicate the phylum or division to which the organism belongs. Microorganisms that you are unable to identify should be sketched in the space provided. It is not necessary to draw those that are identified.

PROTOZOA

GENUS	PHYLUM	BOTTLE NO.	SKETCHES OF UNIDENTIFIED

ALGAE

GENUS	DIVISION	BOTTLE NO.	SKETCHES OF UNIDENTIFIED

CYANOBACTERIA

GENUS	BOTTLE NO.	SKETCHES OF UNIDENTIFIED

B. General Questions

Record the answers to the following questions in the answer column. It may be necessary to consult your text or library references for one or two of the answers.

1. Give the kingdom in which each of the following groups of organisms is found:

   a. protozoans
   b. algae
   c. cyanobacteria
   d. bacteria
   e. fungi
   f. microscopic invertebrates

2. Four kingdoms are represented by the organisms in the above question. Name the fifth kingdom.
3. What is the most significant characteristic seen in eukaryotes that is lacking in prokaryotes?
4. What characteristic in the microscopic invertebrates distinguishes them from protozoans?
5. Which protozoan phylum was not found in pond samples because phylum members are all parasitic?
6. Indicate whether the following are *present* or *absent* in the algae:

   a. cilia   b. flagella   c. chloroplasts

7. Indicate whether the following are *present* or *absent* in the protozoans:

   a. cilia
   b. chloroplasts
   c. mitochondria
   d. mitosis

8. Which photosynthetic pigment is common to all algae and cyanobacteria?
9. Name two photosynthetic pigments that are found in the cyanobacteria but not in the algae.
10. What photosynthetic pigment is found in bacteria but is lacking in all other photosynthetic organisms?
11. What type of movement is exhibited by the diatoms?

Answers

1.a. ______
b. ______
c. ______
d. ______
e. ______
f. ______
2. ______
3. ______
4. ______
5. ______
6.a. ______
b. ______
c. ______
7.a. ______
b. ______
c. ______
d. ______
8. ______
9.a. ______
b. ______
10. ______
11. ______

C. *Protozoan Characterization*

Select the protozoan groups in the right-hand column that have the following characteristics:

1. move with flagella
2. move with cilia
3. move with pseudopodia
4. have nuclear membranes
5. lack nuclear membranes
6. all species are parasitic
7. produce resistant cysts

1. Sarcodina
2. Mastigophora
3. Ciliophora
4. Sporozoa
5. all of above
6. none of above

D. *Characterization of Algae and Cyanobacteria*

Select the groups in the right-hand column that have the following characteristics:

Pigments

1. chlorophyll a
2. chlorophyll b
3. chlorophyll c
4. fucoxanthin
5. c-phycocyanin
6. c-phycoerythrin

1. Euglenophycophyta
2. Chlorophycophyta
3. Chrysophycophyta
4. Phaeophycophyta
5. Pyrrophycophyta
6. Cyanobacteria
7. all of above
8. none of above

Food Storage

7. fats
8. oils
9. starches
10. laminarin
11. leucosin
12. paramylum
13. mannitol

Other Structures

14. pellicle, no cell wall
15. cell walls, box and lid
16. chloroplasts
17. phycobilisomes
18. thylakoids

Answers

Protozoa

1. ______
2. ______
3. ______
4. ______
5. ______
6. ______
7. ______

Algae

1. ______
2. ______
3. ______
4. ______
5. ______
6. ______
7. ______
8. ______
9. ______
10. ______
11. ______
12. ______
13. ______
14. ______
15. ______
16. ______
17. ______
18. ______

LABORATORY REPORT 8, 9

Student: ______________________

Desk No.: __________ Section: __________

Ex. 8 Aseptic Technique

A. Results

1. Were all your transfers successful? ______________________
2. What evidence do you have that they were successful? ______________________
3. What evidence do you have that a transfer is unsuccessful? ______________________

B. Questions:

1. What kinds of organisms are destroyed when your desktop is scrubbed down with a disinfectant?
2. Are bacterial endospores destroyed? ______________________
3. How hot should inoculating loops and needles be heated? ______________________
4. Why is it necessary to flame the mouth of the tube before and after performing an inoculation?

Ex. 9 The Bacteria

A. *Questions:* After you have tabulated your results on the back of this sheet, answer the following questions:

1. Using the number of colonies as an indicator, which habitat sampled by the class appears to be the most contaminated one? ______________________
2. Why do you suppose this habitat contains such a high microbial count? ______________________
3. a. Were any plates completely lacking in colonies? ______________________
   b. Do you think that the habitat sampled was really sterile? ______________________
   c. If your answer to *b* is *no,* then how can you account for the lack of growth on the plate?
   d. If your answer to *b* is *yes,* defend it: ______________________
4. In a few words describe some differences in the macroscopic appearance of bacteria and mold colonies:

Bacteria

B. Tabulation

After examining your TSA and blood agar plates, record your results in the following table and on a similar table that your instructor has drawn on the chalkboard. With respect to the plates, we are concerned with a quantitative evaluation of the degree of contamination and differentiation as to whether the organisms are bacteria or molds. Quantify your recording as follows:

0 no growth
\+ 1 to 10 colonies
\+ + 11 to 50 colonies
\+ + + 51 to 100 colonies
\+ + + + over 100 colonies

After shaking the tube of broth to disperse the organisms, look for cloudiness (turbidity). If the broth is clear, no bacterial growth occurred. Record no growth as 0. If tube is turbid, record + in last column.

STUDENT INITIALS	PLATE EXPOSURE METHOD		COLONY COUNTS		BROTH	
	TSA	Blood Agar	Bacteria	Mold	Source	Result

LABORATORY REPORT 10

Student: ______________________________

Desk No.: __________ Section: ______________

The Fungi:
Yeasts and Molds

A. *Yeast Study*

Draw a few representative cells of *Saccharomyces cerevisiae* in the appropriate circles below. Blastospores (buds) and ascospores, if seen, should be shown and labeled.

Prepared Slide

Living Cells

B. *Mold Study*

In the following table, list the genera of molds identified in this exercise. Under colony description, give the approximate diameter of the colony, its topside color and backside (bottom) color. For microscopic appearance, make a sketch of the organism as it appears on slide preparation.

GENUS	COLONY DESCRIPTION	MICROSCOPIC APPEARANCE (DRAWING)

C. *Questions*

Record the answers for the following questions in the answer column.

1. The science that is concerned with the study of fungi is called ___________________.
2. The kingdom to which the fungi belong is ___________________.
3. Microscopic filaments of molds are called ___________________.
4. A filamentlike structure formed by a yeast from a chain of blastospores is called a ___________________.
5. A mass of mold filaments, as observed by the naked eye, is called a ___________________.
6. Most molds have ___________________ hyphae (*septate* or *nonseptate*).
7. List three kinds of sexual spores that are the basis for classifying the molds.
8. What is the name of the rootlike structure that is seen in *Rhizopus?*
9. What type of hypha is seen in *Mucor* and *Rhizopus?*
10. What kind of asexual spores are seen in *Mucor* and *Rhizopus?*
11. What kind of asexual spores are seen in *Penicillium?*
12. What kind of asexual spores are seen in *Alternaria?*
13. Which subdivision of the Amastigomycota contains individuals that lack sexual spores?
14. What division of Myceteae consists of slime molds?
15. Fungi that exist both as yeasts and molds are said to be _________.

Answers

1. ___________________
2. ___________________
3. ___________________
4. ___________________
5. ___________________
6. ___________________

7.a. ___________________

b. ___________________

c. ___________________

8. ___________________
9. ___________________
10. ___________________
11. ___________________
12. ___________________
13. ___________________
14. ___________________
15. ___________________

LABORATORY REPORT 11–14

Student: ______________________________

Desk No.: __________ Section: ____________

Negative Staining, Smear Preparation, Simple and Capsular Staining

A. Negative Staining (Exercise 11)

1. **Drawing:** Make a drawing in the circle at the right of some of the organisms as seen under oil immersion.
2. In addition to nigrosine, what other agent is often used for making negative-stained slides?

   ____

3. Other than bacteria, what kinds of microorganisms might one encounter in the mouth?

   ____

Oral organisms
(nigrosine)

B. Smear Preparation (Exercise 12)

1. Give two reasons for heating the slide after the smear is air-dried:

   a. ____

   b. ____

2. Why is an inoculating needle preferred to a wire loop when making smears from solid media?

   ____

   ____

   ____

C. Simple Staining (Exercise 13)

1. **Drawing:** Draw a few cells of *C. diphtheriae* from the portion of the slide that exhibits metachromatic granules and palisade arrangement.
2. Why are basic dyes more successful on bacteria than acidic dyes?

   ____

   ____

   ____

Corynebacterium diphtheriae

3. List three basic dyes that are used to stain bacteria: a. ______________________,

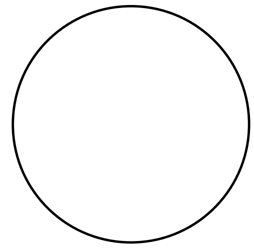
   b. ______________________, c. ______________________

Negative Staining, Smear Preparation, Simple and Capsular Staining

D. Capsular Staining (Exercise 14)

1. List some of the chemical substances that have been identified in bacterial capsules.

2. What relationship is there between capsules and bacterial virulence?



Klebsiella pneumoniae
(capsular stain)

LABORATORY REPORT 15–18

Student: ____________________

Desk No.: __________ Section: __________

Differential Staining: Gram, Spore, and Acid-Fast

A. *Drawings*

With colored pencils, draw the various organisms as seen under oil immersion. Extra circles are provided for additional assignments, if needed.

P. aeruginosa & S. aureus (Gram stain)

B. megaterium & M. B. catarrhalis (Gram stain)

M. smegmatis (Gram stain)

B. megaterium (Schaeffer-Fulton method)

B. megaterium (Dorner method)

M. smegmatis & S. aureus (Ziehl-Neelsen method)

M. phlei & S. aureus (Truant method)

__________________ __________________

OPTIONAL STAINING

Differential Staining: Gram, Spore, and Acid-Fast

B. Completion Questions

1. What color would you expect *S. aureus* to be if the iodine step were omitted in the Gram staining procedure? ______________________________
   Explanation: ______________________________
   ____
   ____

2. What part of the bacterial cell (*cell wall* or *protoplast*) appears to play the most important role in determining whether an organism is gram-positive?
   ____

3. Why would methylene blue not work just as well as safranin for counterstaining in the Gram staining procedure? ______________________________

4. Why are endospores so difficult to stain? ______________________________

5. How do the following two genera of spore-formers differ physiologically?
   Bacillus: ______________________________
   Clostridium: ______________________________

6. How do you differentiate *S. aureus* and *M. B. catarrhalis* from each other on the basis of morphological characteristics? ______________________________
   ____
   ____

7. Are the acid-fast mycobacteria gram-positive or gram-negative? ______________________________

8. For what two diseases is acid-fast staining of paramount importance?
   a. ______________________________ b. ______________________________

9. What advantage does the Ziehl-Neelsen technique have over the Truant method?
   ____
   ____

10. What advantage does the Truant method have over the Ziehl-Neelsen technique?
    ____

11. Why is it desirable to combine *S. aureus* with acid-fast organisms such as *M. smegmatis* when applying an acid-fast staining technique?
    ____
    ____

LABORATORY REPORT 19

Student: ______________________

Desk No.: ________ Section: __________

Motility Determination

A. Test Results

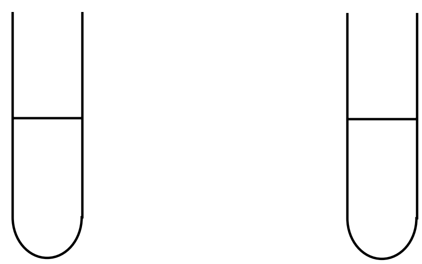
1. Which of the two organisms exhibited true motility on the slides?

____

2. Did the semisolid medium inoculations confirm the results obtained from the slides?

____

3. Sketch in the appearance of the two tube inoculations:



Micrococcus luteus ***Proteus vulgaris***

B. Questions

1. How does Brownian movement differ from true motility? ______________________

____

____

2. How do you differentiate water current movement from true motility?______________

____

____

3. Make sketches that illustrate each of the following flagellar arrangements:

Monotrichic **Lophotrichic**

Amphitrichic **Peritrichic**

LABORATORY REPORT 20

Student: ______________________________

Desk No.: __________ Section: __________

Culture Media Preparation

1. How do the following types of organisms differ in their carbon needs?

   Photoautotrophs: ______________________________

   Photoheterotrophs: ______________________________

2. Where do the above two types of organisms get their energy? ______________________________

3. Where do **chemoheterotrophs** get their energy? ______________________________

4. What is a growth factor? ______________________________

   ____

5. Give two reasons why agar is such a good ingredient for converting liquid media to solid media.

   a. ______________________________

   b. ______________________________

6. Differentiate between the following two types of media:

   Synthetic medium: ______________________________

   Nonsynthetic medium: ______________________________

7. Differentiate between the following two types of media:

   Selective medium: ______________________________

   Differential medium: ______________________________

8. Briefly, list the steps that you would go through to make up a batch of nutrient agar slants:

   ____

   ____

   ____

   ____

LABORATORY REPORT 21

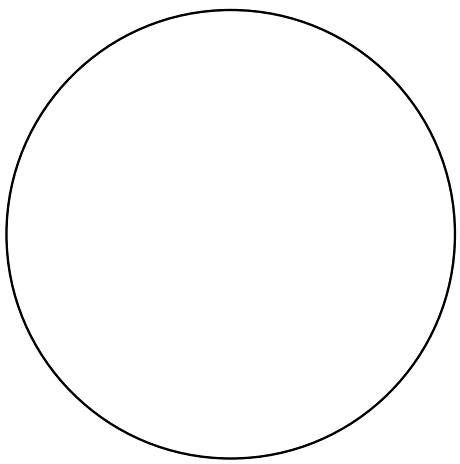
Student: ____________________

Desk No.: ________ Section: ____________

Pure Culture Techniques

A. Evaluation of Streak Plate

Show within the circle the distribution of the colonies on your streak plate. To identify the colonies, use red for *Serratia marcescens,* yellow for *Micrococcus luteus,* and purple for *Chromobacterium violaceum.* If time permits, your instructor may inspect your plate and enter a grade where indicated.



Grade ____________

B. Evaluation of Pour Plates

Show the distribution of colonies on plates II and III, using only the quadrant section for plate II. If plate III has too many colonies, follow the same procedure. Use colors.

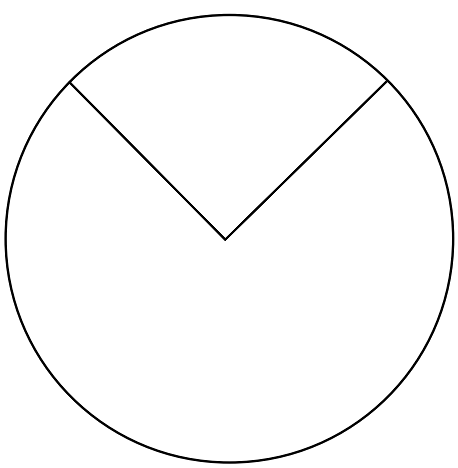


plate II

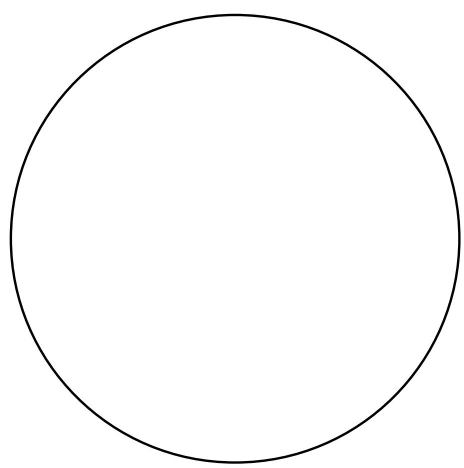
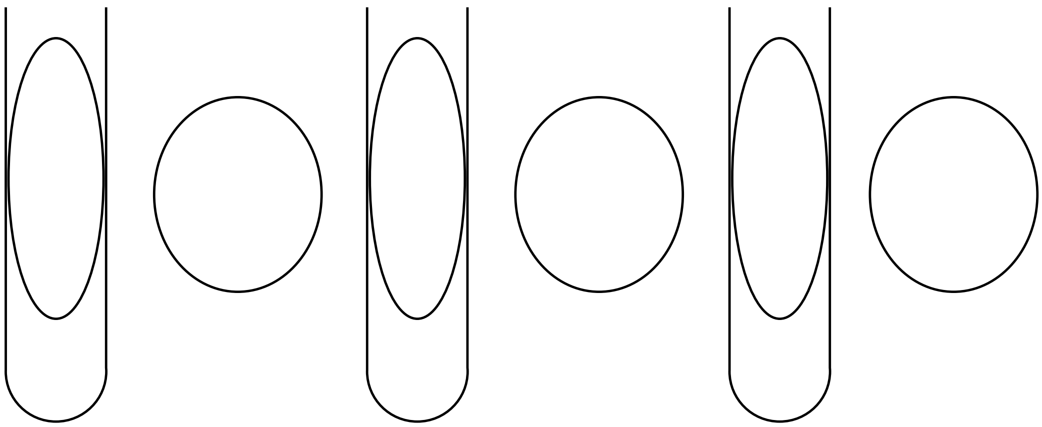


plate III

Pure Culture Techniques

C. Subculture Evaluation

With colored pencils, sketch the appearance of the growth on the slant diagrams below. Also, draw a few cells of each organism as revealed by Gram staining in the adjacent circle.



Serratia marcescens | ***Micrococcus luteus* or *Chromobacterium violaceum*** | ***Escherichia coli***

D. Questions

1. Which method of separating organisms seems to achieve the best separation?
____
2. Which method requires the greatest skill? ______________________________
3. Do you think you have pure cultures of each organism on the slants?______________________________
   Can you be absolutely sure by studying its microscopic appearance?______________________________
   ____
   Explain:______________________________
4. Give two reasons why the nutrient agar must be cooled to 50° C before inoculating and pouring.
____
____
____
5. Why should a Petri plate be discarded if media is splashed up the side to the top?
____
____
6. Give two reasons why it is important to invert plates during incubation:
____
____
7. Why is it important not to dig into the agar with the loop? ______________________________
____
8. Why must the loop be flamed before entering a culture?______________________________
   Why must it be flamed after making an inoculation?______________________________
   ____

LABORATORY REPORT 22

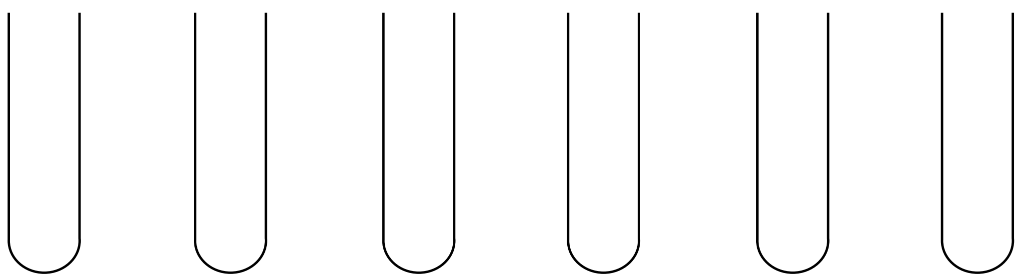
Student: ______________________

Desk No.: ________ Section: ________

Cultivation of Anaerobes

A. Tube Inoculations

After carefully comparing the appearance of the six cultures belonging to you and your laboratory partner, select the best tube for each organism and sketch its appearance in the tubes below. Indicate under each name the type of medium (FTM or TGYA).



E. coli	*S. faecalis*	*S. aureus*	*B. subtilis*	*C. sporogenes*	*C. rubrum*
( )	( )	( )	( )	( )	( )

B. Plate Inoculations

After comparing the growths on the two plates of Brewer's anaerobic agar with the growths in the six tubes, classify each organism as to its oxygen requirements:

Escherichia coli: ____________________

Streptococcus faecalis: ____________________

Staphylococcus aureus: ____________________

Bacillus subtilis: ____________________

Clostridium sporogenes: ____________________

Clostridium rubrum: ____________________

C. Questions

1. What is the function of oxygen at the cellular level? ____________________

2. Why are facultative organisms able to grow with or without oxygen while aerobes grow only in its presence? ____________________

3. How do "indifferents" differ from "facultatives"? ____________________

4. What is the function of the following agents in the media used in this experiment?

   Sodium thioglycollate: ____________________

   Resazurin: ____________________

   Agar in FTM: ____________________

   Agar in TGYA shake: ____________________

Cultivation of Anaerobes

5. How is oxygen removed from the air in a GasPak anaerobic jar? ______________________________

____

____

____

D. Spore Study

If a spore-stained slide is made of the three spore-formers, draw a few cells of each organism in the spaces provided below:

B. subtilis ***C. rubrum*** ***C. sporogenes***

Laboratory Report 23

Student: ______________________

Desk No.: __________ Section: __________

Bacterial Population Counts

A. Quantitative Plating Method

1. Record your plate counts in this table:

DILUTION PLATED	ML PLATED	NUMBER OF COLONIES
1:10,000	1.0	
1:100,000	0.1	
1:1,000,000	1.0	
1:10,000,000	0.1	

2. How many cells per ml were there in the undiluted culture? ______________________
3. How would you inoculate a plate to get 1:100 dilution? ______________________
4. How would you inoculate a plate to get 1:10 dilution? ______________________
5. Give two reasons why it is necessary to shake the water blanks as recommended.
   a. ______________________
   b. ______________________

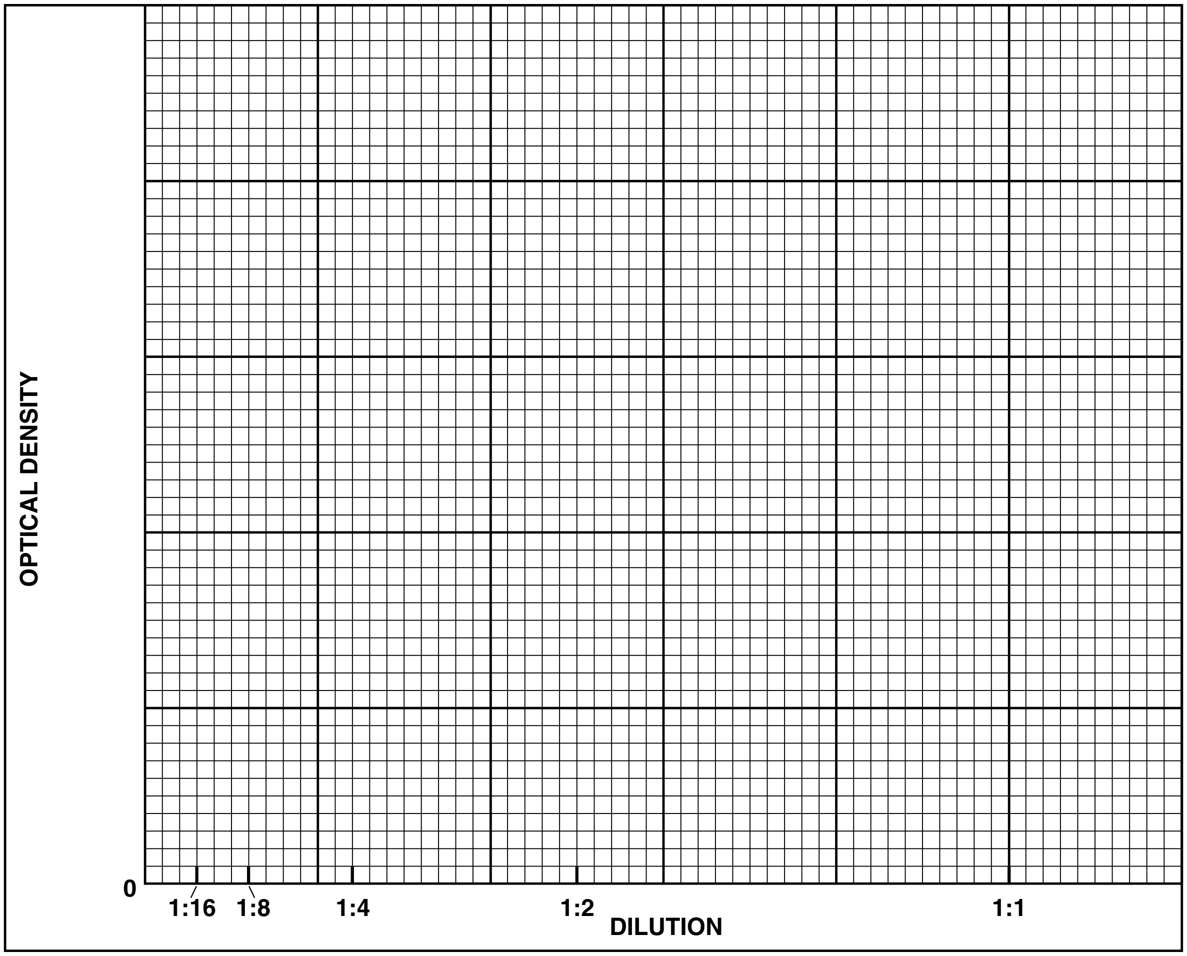
B. Turbidimetric Determinations

1. Record the percent transmittance and optical density values for your dilutions in the following table.

DILUTION	PERCENT TRANSMITTANCE	OPTICAL DENSITY
1:1		
1:2		
1:4		
1:8		
1:16		

Bacterial Population Counts

2. Plot the optical densities versus the concentration of organisms. Complete the graph by drawing a line between plot points.



C. Questions

1. What is the maximum O.D. that is within the linear portion of the curve? ____________________
2. What is the corrected or true O.D. of the undiluted culture? (Hint: If the O.D. for the 1:2 dilution but not the 1:1 dilution is within the linear portion of the curve, then the O.D. of the 1:1 dilution should not be considered correct. The correct or true O.D. of the undiluted culture in this example could be estimated by multiplying the O.D. of the 1:2 dilution by 2.) ____________________
3. What is the correlation between corrected O.D. and cell number for your culture? ____________________

   ____

   ____
4. Why is it necessary to perform a plate count in conjunction with the turbidimetry procedure?

   ____
5. If your medium were pale blue instead of amber-colored, as is the case of nutrient broth, would you set the wavelength control knob higher or lower than 686 nanometers?____________________

Laboratory Report
24, 25

Student: ______________________

Desk No.: __________ Section: __________

Ex. 24 Slide Culture:
Autotrophs

A. Microscopic Examination

While examining the two slides, move them around to different areas to note the various types of organisms that are present. Draw representative types.

GRAM'S STAIN **LIVING**

B. Questions

1. With respect to Gram's stain, which type (*gram-positive* or *gram-negative*) seems to predominate?

   ____

2. List as many different kinds of autotrophic protists as you can that can be cultured on this type of slide.

   ____

   ____

   ____

3. Some organisms that grow on this type of slide are chemosynthetic heterotrophs. What would be the source of their food?

   ____

   ____

Ex. 25 Slime Mold Culture

A. Observations

1. What happened when the flow of protoplasm on a plasmodium was interrupted by severance with a scalpel?

____

____

2. Describe your observations of the crushed spores on the hanging drop slide.

____

____

____

B. Questions

1. List two functions served by fructification (sporangia formation) in *Physarum.*

   a. ____

   b. ____

2. What is the principal function of the plasmodial stage of *Physarum?*

____

3. List two characteristics that the Myxobacterales and Gymnomycota have in common.

   a. ____

   b. ____

4. Postulate as to the evolutionary relationship between the myxobacteria and slime molds.

____

____

____

____

____

____

LABORATORY REPORT 27

Student: ____________________

Desk No.: __________ Section: __________

Anaerobic Phototrophic Bacteria: Isolation and Culture

A. Observations:

1. **Column Appearance:** Describe in a few words the appearance of the Winogradsky column during the:

   First Period:

   ____

   One Week Later:

   ____

   Two Weeks Later:

   ____

   Subsequent Weeks:

   ____

   ____

2. **Subculture Appearance:** Describe the appearance of your subculture tubes at the end of two weeks:

   ____

   ____

   ____

3. **Microscopic Appearance:** Describe the characteristics of the cells you observed on wet mount slides and gram-stained slides: ________________________________

   ____

B. Questions:

1. What roles do the following organisms perform in the Winogradsky column?

   Clostridium: ________________________________

   Desulfovibrio: ________________________________

2. How do *Chromatium* and *Chlorobium* differ as to where they put the sulfur that they produce? __________

   ____

3. Give the equations for photosynthesis in the following:

   Algae: ________________________________

   Chromatium* and *Chlorobium: ________________________________

Anaerobic Phototrophic Bacteria: Isolation and Culture

C. Tabulation of Results

If different mud samples were used by class members, record your results on the chart below and on a similar chart constructed by your instructor on the chalkboard. Record the presence (+) or absence (−) of the various species identified. In the "Other" column list any other species that you might have encountered.

Student Initials	Source of Pond Mud	*Chromatium*	*Chlorobium*	Other

D. Conclusions from above table:

LABORATORY REPORT 28, 29

Student: ____________________

Desk No.: __________ Section: __________

Isolation of Phage from Sewage and Flies

A. *Plaque Size Increase (for both Exercises 28 and 29)*

With a china marking pencil, circle and label three plaques on one of the plates and record their sizes in millimeters at 1-hour intervals.

TIME	PLAQUE SIZE (millimeters)		
	Plaque No. 1	Plaque No. 2	Plaque No. 3
When first seen			
1 hour later			
2 hours later			
3 hours later			

B. *Questions (for Exercise 28 only)*

1. Were any plaques seen on the negative control plate? ____________________
2. Do plates 1, 2, and 3 show a progressive increase in number of plaques with increased amount of sewage filtrate? ____________________
3. Did the phage completely "wipe out" all bacterial growth on any of the plates? ____________________

   If so, which plates? ____________________

C. *Observations (for Exercise 29 only)*

Count all the plaques on each plate and record the counts in the following table. If the plaques are very numerous, use a Quebec colony counter and hand counting device. If this exercise was performed as a class project with individual students doing only one or two plates from a common fly-broth filtrate, record all counts on the chalkboard on a table similar to the one below.

Plate Number	1	2	3	4	5	6	7	8	9	10
E. coli (ml)	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	1.0
Filtrate (ml)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0
Number of plaques										

Isolation of Phage from Sewage and Flies

D. Questions (for Exercise 29 only)

1. Which plate was used as the negative control? ______________________________
2. Were there any plaques on the negative control plate?______________________________
3. What would be the explanation for the presence of plaques on the negative control plate?

   ____

   ____
4. Were any plates completely "wiped out" by phage action? ______________________________

   If so, which ones?______________________________

E. Terminology (for both Exercises 28 and 29)

1. Differentiate between the following:

   Lysis: ______________________________

   Lysogeny:______________________________
2. Differentiate between the following:

   Virulent phage: ______________________________

   Temperate phage: ______________________________

LABORATORY REPORT 30

Student: ______________________________

Desk No.: __________ Section: ____________

Burst Size Determination: A One-Step Growth Curve

A. *Plaque Counts*

Record the counts of plaques on each of the plates in the following table. Record the peak number of plaques as the *burst size*. The drop in plaque numbers after a peak results from adsorption of mature phage virions on other bacterial cells and cell debris.

15	25	30	35	40	45	50

Burst size: __________

B. *Dilution Interpretation*

Answer the following questions to clarify your understanding of the dilutions that occur in this experiment.

1. How many cells were present in each milliliter of the original bacterial culture? __________
2. How many bacterial cells (total) were dispensed into the ADS tube? __________
3. If the bacterial dilution per plate is 1:10,000,000, how many bacterial cells were distributed to each plate?

   ____

4. How many phage virions were present in 1 ml of the original phage suspension?

   ____

5. How many phage virions were present in the 0.1 ml of phage suspension that was added to the ADS tube?

   ____

6. What was the numerical ratio of phage virions to bacterial cells in the ADS tube? __________

   What is this ratio called? __________
7. How many bacterial cells were placed in the ADS-2 tube? __________
8. What effect does dilution have on adsorption? __________

LABORATORY REPORT 31–33

Student: ______________________

Desk No.: __________ Section: __________

Microbial Interrelationships
Ex. 31 Bacterial Commensalism

A. Results

Indicate the degree of turbidity (none, +, + +, + + +) in the following table. With colored pencils, draw the appearance of the gram-stained slides where indicated.

ORGANISMS	TURBIDITY	GRAM STAIN
Staphylococcus aureus		
Clostridium sporogenes		
S. aureus and *C. sporogenes*		

B. Questions

1. Does *C. sporogenes* grow well in nutrient broth?______________________
2. How does *S. aureus* assist *C. sporogenes* in growth?______________________

____

Ex. 32 Bacterial Synergism

A. Results

Examine the six tubes of media, looking for acid and gas. In the presence of acid, bromthymol blue turns yellow. Record your results in the table below. Consult other students for their results and complete the table.

INDIVIDUAL ORGANISMS	LACTOSE		SUCROSE		COMBINATIONS	LACTOSE		SUCROSE	
	Acid	Gas	Acid	Gas		Acid	Gas	Acid	Gas
E. coli					*E. coli* and *P. vulgaris*				
P. vulgaris					*E. coli* and *S. aureus*				
S. aureus					*S. aureus* and *P. vulgaris*				

Bacterial Synergism and Microbial Antagonism

B. Questions

1. Did any of the three organisms produce gas in either lactose or sucrose broth when alone in the medium?

2. Which organisms act synergistically to produce gas in

   lactose? ______

   sucrose? ______

Ex. 33 Microbial Antagonism

A. Questions

1. Which organisms are antagonistic to *E. coli?* ______

2. Which organisms are antagonistic to *S. aureus?* ______

3. Name an antibiotic substance that is derived from each of the following types of organisms. Also, indicate whether the substance is effective against gram-positive or gram-negative organisms.

   A bacterium: ______

   An actinomyces: ______

   A fungus: ______

4. What role does microbial antagonism play in nature? ______

5. In what physiological way does penicillin affect penicillin-sensitive bacteria? ______

6. How do sulfonamides inhibit bacterial growth? ______

LABORATORY REPORT 34

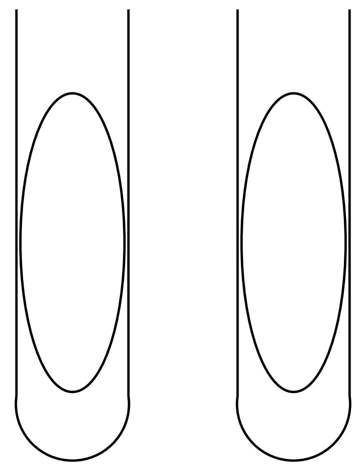
Student: ______________________

Desk No.: __________ Section: __________

Temperature: Effects on Growth

A. Pigment Formation and Temperature

1. Draw the appearance of the growth of *Serratia marcescens* on the nutrient agar slants using colored pencils.
2. Which temperature seems to be closest to the optimum temperature for pigment formation? ______________________
3. What are the cellular substances that control pigment formation and are regulated by temperature? ______________________



25° C   38° C

B. Growth Rate and Temperature

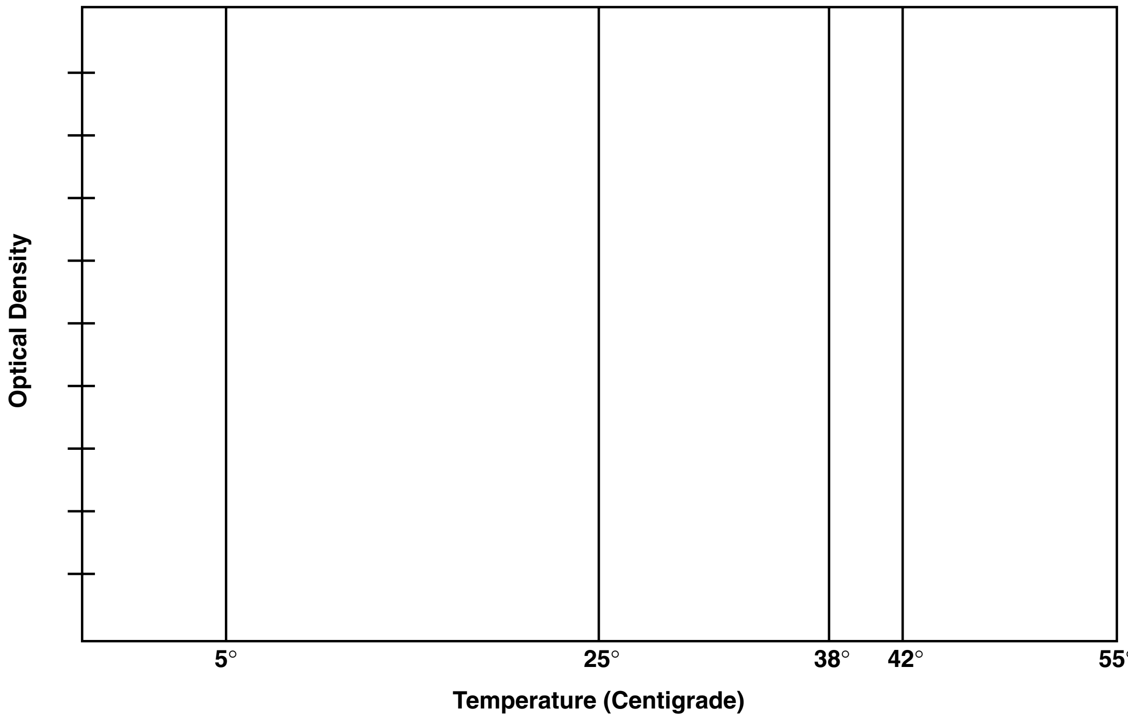
If a spectrophotometer is available, dispense the cultures into labeled cuvettes and determine the percent transmittance of each culture. Calculate the O.D. values from the percent transmittance, using the formula given in Exercise 23.

If no spectrophotometer is available, record only the visual reading as +, + +, + + +, and none.

Temp. °C	SERRATIA MARCESCENS			ESCHERICHIA COLI		
	Visual Reading	Spectrophotometer		Visual Reading	Spectrophotometer	
		%T	O.D.		%T	O.D.
5						
25						
38						
42						
55						

Temperature: Effects on Growth

Growth curves of *Serratia marcescens* and *Escherichia coli* as related to temperature.



1. On the basis of the above graph, estimate the optimum growth temperature of the two organisms.

   Serratia marcescens: ______________________________

   Escherichia coli: ______________________________

2. To get more precise results for the above graph, what would you do?

   ____

   ____

3. Differentiate between the following:

   Thermophile: ______________________________

   Mesophile: ______________________________

   Psychrophile: ______________________________

4. What is the optimum growth temperature range for most psychrophiles? ______________________________

Laboratory Report 35

Student: ____________________

Desk No.: ________ Section: ________

Temperature: Lethal Effects

A. Tabulation of Results

Examine your five Petri plates, looking for evidence of growth. Record on the chalkboard, using a chart similar to the one below, the presence or absence of growth as (+) or (−). When all members of the class have recorded their results, complete this chart.

ORGANISM	60° C					70° C					80° C					90° C					100° C				
	C*	10	20	30	40	C*	10	20	30	40	C*	10	20	30	40	C*	10	20	30	40	C*	10	20	30	40
S. aureus																									
E. coli																									
B. megaterium																									

*C = control tube

1. If they can be determined from the above information, record the **thermal death point** for each of the organisms.

   S. aureus: ____________ *E. coli:* ____________ *B. megaterium:* ____________

2. From the following table, determine the **thermal death time** for each organism at the tabulated temperatures.

ORGANISM	THERMAL DEATH TIME				
	60° C	70° C	80° C	90° C	100° C
S. aureus					
E. coli					
B. megaterium					

B. Questions

1. Give three reasons why endospores are much more resistant to heat than are vegetative cells.

   a. ____________________

   b. ____________________

   c. ____________________

Temperature: Lethal Effects

2. Differentiate between the following:

   Thermoduric: ______________________________

   ____

   Thermophilic: ______________________________

   ____

3. List four diseases caused by spore-forming bacteria.

   a. ______________ b. ______________

   c. ______________ d. ______________

4. Since boiling water is unreliable in destroying endospores, how should one use heat in medical applications to ensure spore destruction? (three ways)

   a. ______________________________

   b. ______________________________

   c. ______________________________

LABORATORY REPORT 36

Student: ____________________

Desk No.: ________ Section: __________

pH and Microbial Growth

A. Tabulation of Results

If a spectrophotometer is available, dispense the cultures into labeled cuvettes and determine the percent transmittance of each culture. Calculate the O.D. values from the percent transmittance, using the formula given in Exercise 23. To complete the tables, get the results of the other three organisms from other members of the class, and delete the substitution organisms in the tables that were not used.

If no spectrophotometer is available, record only the visual reading as +, + +, + + +, and none.

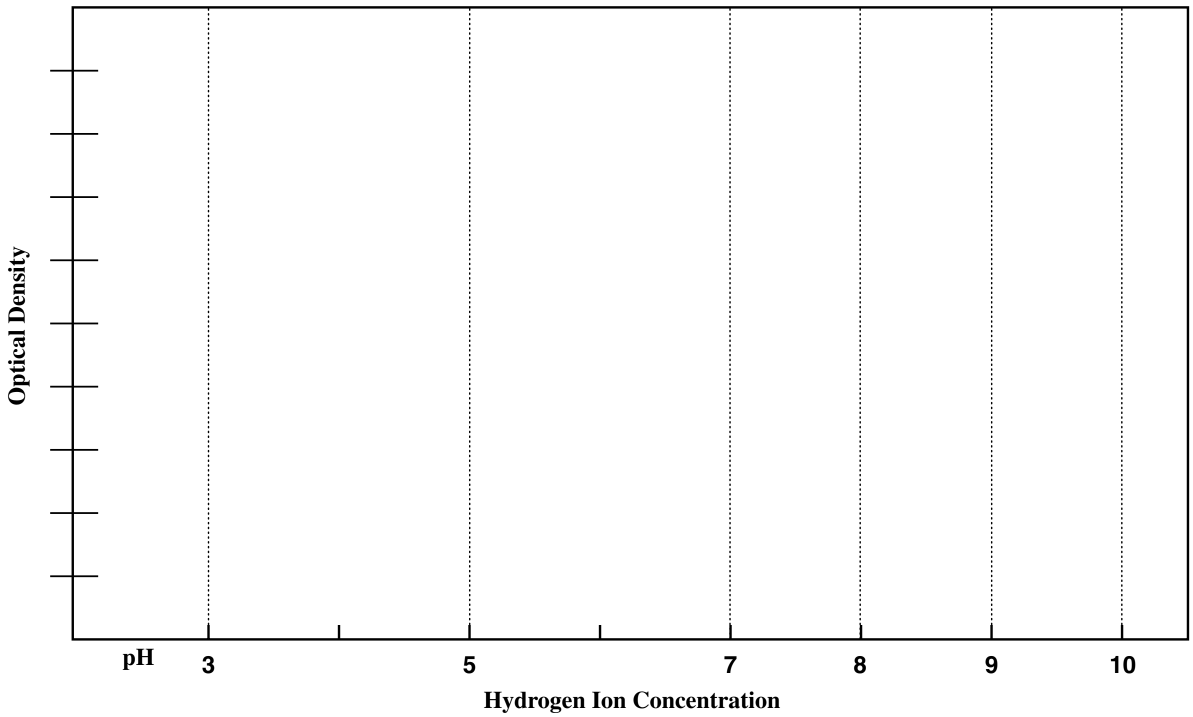
pH	*Escherichia coli*			*Staphylococcus aureus*		
	Visual Reading	Spectrophotometer		Visual Reading	Spectrophotometer	
		%T	O.D.		%T	O.D.
3						
5						
7						
8						
9						
10						

pH	*Alcaligenes faecalis or Sporosarcina ureae*			*Saccharomyces cervisiae or Candida glabrata*		
	Visual Reading	Spectrophotometer		Visual Reading	Spectrophotometer	
		%T	O.D.		%T	O.D.
3						
5						
7						
8						
9						
10						

pH and Microbial Growth

B. Growth Curves

Once you have computed all the O.D. values on the two tables, plot them on the following graph. Use different colored lines for each species.



C. Questions

1. Which organism seems to grow best in acid media? ______________________
2. Which organism seems to grow best in alkaline media? ______________________
3. Which organism seems to tolerate the broadest pH range? ______________________

LABORATORY REPORT 37, 38

Student: ______________________

Desk No.: __________ Section: __________

Ex. 37 Osmotic Pressure and Bacterial Growth

A. Results

Record the amount of growth of each organism at the different salt concentrations, using, +, ++, +++, and none to indicate degree of growth.

ORGANISM	SODIUM CHLORIDE CONCENTRATION							
	0.5%		5%		10%		15%	
	48 hr	96 hr	48 hr	96 hr	48 hr	96 hr	48 hr	96 hr
Escherichia coli								
Staphylococcus aureus								
Halobacterium salinarium								

B. Questions

1. Evaluate the salt tolerance of the above organisms.

   Tolerates very little salt: ______________________

   Tolerates a broad range of salt concentration: ______________________

   Grows only in the presence of high salt concentration: ______________________

2. How would you classify *Halobacterium salinarium* as to salt needs? Check one.

   Obligate halophile __________ Facultative halophile __________

3. Differentiate between the following:

   Halophile: ______________________

   Osmophile: ______________________

4. Supply the following information concerning **mannitol salt agar** (Refer to the Difco Manual).

   Composition: ______________________

   ____

   For what organism is this medium selective? ______________________

   What ingredient makes it selective? ______________________

Oligodynamic Action

Ex. 38 Oligodynamic Action

A. Tabulation of Results

Measure the zone of inhibition from the edge of each piece of metal with a millimeter ruler. Record the measurements in the table. Spaces are provided for write-in of additional metals.

METAL	MILLIMETERS OF INHIBITION	
	E. coli	*S. aureus*
Copper		
Silver		
Aluminum		

B. Questions

1. Which metal seems to exhibit the greatest amount of oligodynamic action?

   ____

2. Which metal or metals seem to be ineffective? ______________________________

3. Do these two organisms seem to differ in their susceptibility to oligodynamic action?

   ____

   Explain: ______________________________

   ____

   ____

4. What specific chemical substances in bacterial cells are inactivated by heavy metals, affecting growth?

   ____

   ____

LABORATORY REPORT 39

Student: ____________________

Desk No.: __________ Section: __________

Ultraviolet Light:
Lethal Effects

A. Tabulation of Results

Your instructor will construct a table similar to the one below on the chalkboard for you to record your results. If substantial growth is present in the exposed area, record your results as +++. If three or fewer colonies survived, record +. Moderate survival should be indicated as ++. No growth should be recorded as −. Record all information in the table.

ORGANISMS	EXPOSURE TIMES							
S. aureus	10 sec	20 sec	40 sec	80 sec	2.5 min	5 min	10 min	• 20 min
Survival								
B. megaterium	1 min	2 min	4 min	8 min	15 min	30 min	60 min	• 6 min
Survival								

*Plates covered during exposure.

B. Questions

1. What length of time is required for the destruction of non-spore-forming bacteria such as *Staphylococcus aureus?*

2. Can you express, quantitatively, how much more resistant *B. megaterium* spores are to ultraviolet light than *S. aureus* vegetative cells (i.e., how many *times* more resistant are they)?

3. Why is it desirable to remove the cover from the Petri dish when making exposures?

4. In what specific way does ultraviolet light destroy microorganisms?

5. What adverse effect can result from overexposure of human tissues to ultraviolet light?

Ultraviolet Light: Lethal Effects

6. What wavelength of ultraviolet is most germicidal? ______________________________
7. List several practical applications of ultraviolet light to microbial control.

____
____
____
____
____
____
____
____

LABORATORY REPORT 40

Student: ____________________

Desk No.: __________ Section: __________

Evaluation of Disinfectants: The Use-Dilution Method

A. *Tabulation of Results*

The instructor will draw a table on the chalkboard similar to the one below. Examine your tubes of nutrient broth and pins by shaking them and looking for growth (turbidity). If you are doubtful as to whether growth is present, compare the tubes with a tube of sterile nutrient broth. Record on the chalkboard a plus (+) sign if growth is present and a minus (−) sign if no growth is visible. After all students have recorded their results, complete the following chart.

DISINFECTANT		MINUTES											
		Staphylococcus aureus						*Bacillus megaterium*					
	Substitution	C*	1	5	10	30	60	C*	1	5	10	30	60
1:750 Zephiran													
5% phenol													
8% formaldehyde													

*C = control tube

B. *Questions*

1. What conclusions can be drawn from this experiment?

____

____

____

____

____

____

____

____

____

____

2. Distinguish between the following:

Disinfectant: ____

____

Antiseptic: ____

____

Evaluation of Disinfectants: The Use-Dilution Method

3. What factors other than time influence the action of a chemical agent on bacteria?

4. Fill in the equation that explains how the **phenol coefficient** is determined:

   P.C. =

5. What are some drawbacks that one encounters when attempting to apply the phenol coefficient to all disinfectants?

LABORATORY REPORT 41

Student: ______________________

Desk No.: ________ Section: __________

Evaluation of Alcohol: Its Effectiveness as a Skin Degerming Agent

A. Tabulation of Results

Count the number of colonies that appear on each of the thumbprints and record them in the following table. If the number of colonies has increased in the second press, record a 0 in percent reduction. Calculate the percentages of reduction and record these data in the appropriate column. Use this formula:

$$\text{Percent Reduction} = \frac{(\text{Colony Count 1st press}) - (\text{Colony Count 2nd press})}{(\text{Colony count 1st press})} \times 100$$

LEFT THUMB (Control)			RIGHT THUMB (Dipped)			RIGHT THUMB (Swabbed)		
Colony Count 1st Press	Colony Count 2nd Press	Percent Reduction	Colony Count 1st Press	Colony Count 2nd Press	Percent Reduction	Colony Count 1st Press	Colony Count 2nd Press	Percent Reduction
Av. % Reduction, Left (C)			Av. % Reduction, Right (D)			Av. % Reduction, Right (S)		

Evaluation of Alcohol: Its Effectiveness as a Skin Degerming Agent

B. Questions

1. In general, what effect does alcohol have on the level of skin contaminants?______________________

   ____

2. Is there any difference between the effects of dipping versus swabbing? ________________________

   Which method appears to be more effective?____

3. There is definitely survival of some microorganisms even after alcohol treatment. Without staining or microscopic scrutiny, predict what types of microbes are growing on the medium where you made the right thumb impression after treatment. ____

   ____

   ____

   ____

   ____

LABORATORY REPORT 42

Student: ____________________

Desk No.: __________ Section: __________

Evaluation of Antiseptics:
The Filter Paper Disk Method

A. Tabulation of Results

With a millimeter scale, measure the zones of inhibition between the edge of the filter paper disk and the organisms. Record this information. Exchange your plates with other students' plates to complete the measurements for all chemical agents.

DISINFECTANT	MILLIMETERS OF INHIBITION	
	Staphylococcus aureus	*Pseudomonas aeruginosa*
5% phenol		
5% formaldehyde		
5% iodine		

B. Questions

1. What conclusions can be derived from these results? ____________________

2. What factors influence the size of the zone of inhibition? ____________________

LABORATORY REPORT 43

Student: ____________________

Desk No.: ________ Section: __________

Antimicrobic Sensitivity Testing: The Kirby-Bauer Method

A. Tabulation

List the antimicrobics that were used for each organism. Consult tables 43.1 and 43.2 to identify the various disks. After measuring and recording the zone diameters, consult table VII in Appendix A for interpretation. Record the degrees of sensitivity (R, I, or S) in the sensitivity column. Exchange data with other class members to complete the entire chart.

	ANTIMICROBIC	ZONE DIA.	RATING (R, I, S)	ANTIMICROBIC	ZONE DIA.	RATING (R, I, S)
S. aureus						
P. aeruginosa						
Proteus vulgaris						
E. coli						

Antimicrobic Sensitivity Testing: The Kirby-Bauer Method

B. Questions

1. Which antimicrobics would be suitable for the control of the following organisms?

   S. aureus: ______

   E. coli: ______

   P. vulgaris: ______

   P. aeruginosa: ______

2. Differentiate between the following:

   Narrow spectrum antibiotic: ______

   Broad spectrum antibiotic: ______

3. Which antimicrobics used in this experiment would qualify as being excellent broad spectrum antimicrobics?

4. Differentiate between the following:

   Antibiotic: ______

   Antimicrobic: ______

5. How can drug resistance in microorganisms be circumvented? ______

LABORATORY REPORT 44

Student: ______________________

Desk No.: ________ Section: __________

Effectiveness of Hand Scrubbing

A. Tabulation of Results

The instructor will draw a table on the chalkboard similar to the one below. Examine the six plates that your group inoculated from the basin of water. Select the two plates of a specific dilution that have approximately 30 to 300 colonies and count all of the colonies of each plate with a Quebec colony counter. Record the counts for each plate and their averages on the chalkboard. Once all the groups have recorded their counts, record the dilution factors for each group in the proper column. To calculate the organisms per milliliter multiply the average count by the dilution factor.

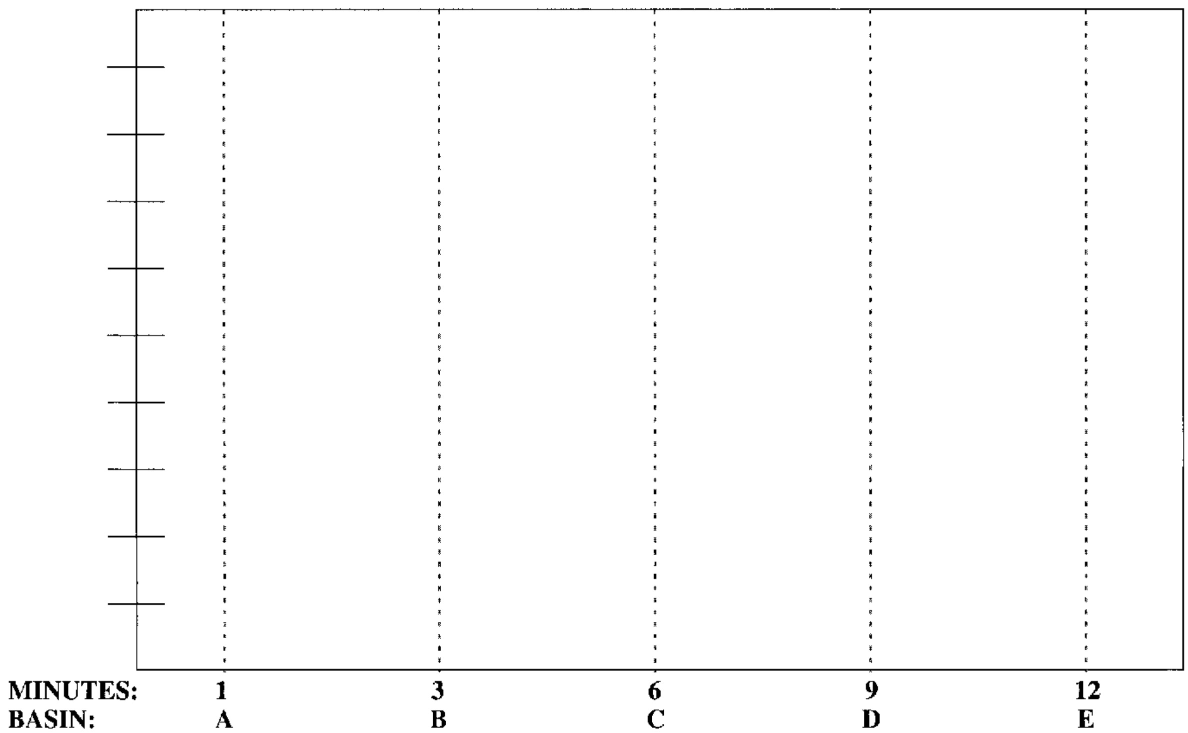
GROUP	0.1 ml COUNT		0.2 ml COUNT		0.4 ml COUNT		DILUTION FACTOR*	ORGANISMS PER MILLILITER
	Per Plate	Average	Per Plate	Average	Per Plate	Average		
A								
B								
C								
D								
E								

*Dilution factors: 0.1 ml = 10; 0.2 ml = 5; 0.4 ml = 2.5

Effectiveness of Hand Scrubbing

B. Graph

After you have completed this tabulation, plot the number of organisms per milliliter that were present in each basin.



C. Questions

1. What conclusions can be derived from this exercise?

2. What might be an explanation of a higher count in Basin D than in B, ruling out contamination or faulty techniques?

3. Why is it so important that surgeons scrub their hands prior to surgery even though they wear rubber gloves?

LABORATORY REPORT 45

Student: ______________________

Desk No.: __________ Section: __________

Preparation and Care of Stock Cultures

1. Why shouldn't cultures be stored at room temperature or in the incubator for any length of time?

2. Why should stock cultures be reinoculated to new media ("rotated") even if they are stored in the refrigerator? ______________________

3. For what types of inoculations do you use your

   reserve stock culture? ______________________

   working stock culture? ______________________

4. What is **lyophilization?** ______________________

   What advantage does this procedure have over the method we are using for maintaining stock cultures?

Laboratory Report 48–50

Student: ____________________

Desk No.: __________ Section: __________

Physiological Characteristics of Bacteria

A. Media

List the media that are used for the following tests:

1. Butanediol production
2. Hydrogen sulfide production
3. Indole production
4. Starch hydrolysis
5. Urease production
6. Citrate utilization
7. Fat hydrolysis
8. Casein hydrolysis
9. Catalase production
10. Mixed acid fermentation
11. Glucose fermentation
12. Nitrate reduction

B. Reagents

Select the reagents that are used for the following tests:

1. Indole test
2. Voges-Proskauer test
3. Catalase test
4. Starch hydrolysis

Barritt's reagent—1
Gram's iodine—2
Hydrogen peroxide—3
Kovacs' reagent—4
None of these—5

C. Ingredients

Select the ingredients of the reagents for the following tests. Consult Appendix B. More than one ingredient may be present in a particular reagent.

1. Oxidase test
2. Voges-Proskauer test
3. Indole test
4. Nitrite test

α-naphthol—1
Dimethyl-α-naphthylamine—2
Dimethyl-ρ-phenylenediamine hydrochloride—3
ρ-dimethylamine benzaldehyde—4
Potassium hydroxide—5
Sulfanilic acid—6

D. Enzymes

What enzymes are involved in the following reactions?

1. Urea hydrolysis
2. Hydrogen gas production from formic acid
3. Casein hydrolysis
4. Indole production
5. Nitrate reduction
6. Starch hydrolysis
7. Fat hydrolysis
8. Gelatin hydrolysis (Ex. 47)
9. Hydrogen sulfide production

Answers

Media

1. ____
2. ____
3. ____
4. ____
5. ____
6. ____
7. ____
8. ____
9. ____
10. ____
11. ____
12. ____

Reagents	Ingredients
1. ____	1. ____
2. ____	2. ____
3. ____	3. ____
4. ____	4. ____

Enzymes

1. ____
2. ____
3. ____
4. ____
5. ____
6. ____
7. ____
8. ____
9. ____

Physiological Characteristics of Bacteria

E. Test Results

Indicate the appearance of the following positive test results.

1. Glucose fermentation, no gas
2. Citrate utilization
3. Urease production
4. Indole production
5. Acetoin production
6. Hydrogen sulfide production
7. Coagulation of milk
8. Peptonization in milk
9. Litmus reduction in milk
10. Nitrate reduction
11. Catalase production
12. Casein hydrolysis
13. Fat hydrolysis

Answers

1. ______
2. ______
3. ______
4. ______
5. ______
6. ______
7. ______
8. ______
9. ______
10. ______
11. ______
12. ______
13. ______

F. General Questions

1. Differentiate between the following:

   Respiration: ______

   Fermentation: ______

   Oxidation: ______

   Reduction: ______

   Catalase: ______

   Peroxidase: ______

2. List two or three difficulties one encounters in trying to differentiate bacteria on the basis of physiological characteristics. ______

3. Now that you have determined the morphological, cultural, and physiological characteristics of your unknown, what other kinds of tests might you perform on the organism to assist in identification?

LABORATORY REPORT 52

Student: ____________________

Desk No.: __________ Section: __________

Enterobacteriaceae Identification: The API 20E System

A. Tabulation of Results

By referring to charts I and II, Appendix D, determine the results of each test and record these results as positive (+) or negative (−) in the table below. Note that the results of the oxidase test must be recorded in the last column on the right side of the table.

ONPG	ADH	LDC	ODC	CIT	H_2S	URE	TDA	IND	VP	GEL	GLU	MAN	INO	SOR	RHA	SAC	MEL	AMY	ARA	OXI
1	2	4	1	2	4	1	2	4	1	2	4	1	2	4	1	2	4	1	2	4
	☐			☐			☐			☐			☐			☐			☐	

NO_2	N_2 GAS	MOT	MAC	OF-O	OF-F
1	2	4	1	2	4
	☐			☐	

Additional Digits

B. Construction of Seven-Digit Profile

Note in the above table that each test has a value of 1, 2, or 4. To compute the seven-digit profile for your unknown, total up the positive values for each group.

Example:
5 144 572 = *E. coli*

ONPG	ADH	LDC	ODC	CIT	H_2S	URE	TDA	IND	VP	GEL	GLU
+	−	+	+	−	−	−	−	+	−	−	+
1	0	4	1	0	0	0	0	4	0	0	4
	5			1			4			4	

MAN	INO	SOR	RHA	SAC	MEL	AMY	ARA	OXI
+	−	+	+	+	+	−	+	−
1	0	4	1	2	4	0	2	0
	5			7			2	

Enterobacteriaceae Identification: The API 20E System

C. Using the API 20E Analytical Index or the API Characterization Chart

If the *API 20E Analytical Index* is available on the demonstration table, use it to identify your unknown, using the seven-digit profile number that has been computed. If no *Analytical Index* is available, use Characterization Chart III in Appendix D.

Name of Unknown: ______________________________

D. Additional Tabulation Blank

If you need another form, use the one below:

api® 20E

Reference Number ____________ Patient ______________________ Date ____________

Source/Site ____________ Physician ______________________ Dept./Service ____________

	ONPG 1	ADH 2	LDC 4	ODC 1	CIT 2	H_2S 4	URE 1	TDA 2	IND 4	VP 1	GEL 2	GLU 4	MAN 1	INO 2	SOR 4	RHA 1	SAC 2	MEL 4	AMY 1	ARA 2	OXI 4
5 h																					
24 h																					
48 h																					
Profile Number																					

	NO_2 1	N_2 GAS 2	MOT 4	MAC 1	OF-O 2	OF-F 4
5 h						
24 h						
48 h						
Additional Digits						

Additional Information

Identification

00-42-012 E-3 (7/80)

E. Questions

1. What is the intended function of the API 20E system? ______________________________

2. In the "real world" who would use this system? ______________________________

3. What might be an explanation for the failure of this system to work with some of the bacterial cultures we use? ______________________________

Laboratory Report 53

Student: ______________________

Desk No.: __________ Section: ____________

Enterobacteriaceae Identification: The Enterotube II System

A. *Tabulation of Results*

Record the results of each test in the following table with a plus (+) or minus (−).

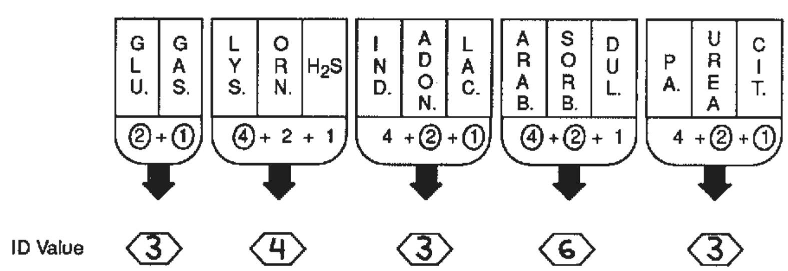
GLUCOSE	GAS PRODUCTION	LYSINE	ORNITHINE	H_2S	INDOLE	ADONITOL	LACTOSE	ARABINOSE	SORBITOL	VOGES-PROSKAUER	DULCITOL	PHENYLALANINE DEAMINASE	UREA	CITRATE

B. *Identification by Chart Method*

If no *Interpretation Guide* is available, apply the above results to chart IV, Appendix D, to find the name of your unknown. Note that the spacing of the above table matches the size of the spaces on chart IV. If this page is removed from the manual, folded, and placed on chart IV, the results on the above table can be moved down the chart to make a quick comparison of your results with the expected results for each organism.

C. *Using the Enterotube II Interpretation Guide*

If the *Interpretation Guide* is available, determine the five-digit code number by circling the numbers (4, 2, or 1) under each test that is positive, and then totaling these numbers within each group to form a digit for that group. Note that there are two tally charts on the next page of this Laboratory Report for your use.



The "ID Value" 34363 can be found by thumbing the pages of the *Interpretation Guide.* The listing is as follows:

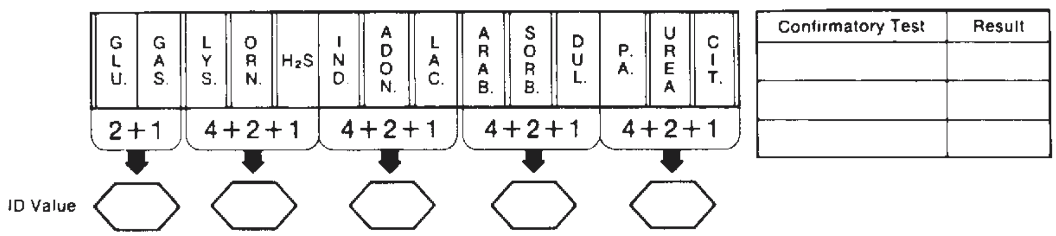
ID Value	**Organism**	**Atypical Test Results**
34363	*Klebsiella pneumoniae*	None

Conclusion: Organism was correctly identified as *Klebsiella pneumoniae.* In this case, the identification was made independent of the V-P test.

Enterobacteriaceae Identification: The Enterotube II System

D. Tally Charts

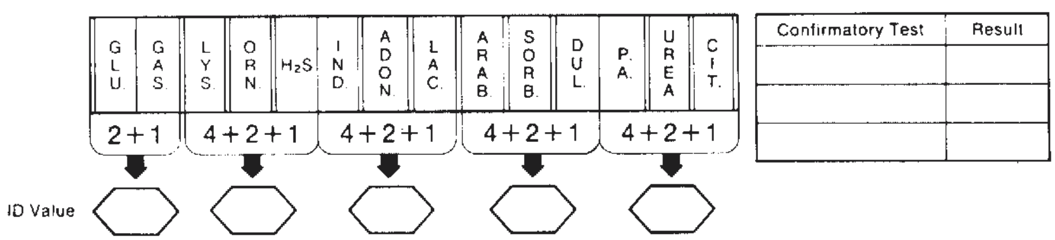
ENTEROTUBE® II*



Culture Number, Case Number or Patient Name ______ Date ______ Organism Identified ______

*VP utilized as confirmatory test only.

ENTEROTUBE® II*



Culture Number, Case Number or Patient Name ______ Date ______ Organism Identified ______

*VP utilized as confirmatory test only.

E. Questions

1. What is the intended function of the Enterotube II System? ______

2. In the "real world" who would use this system? ______

3. What might be an explanation for the failure of this system to work with some of the bacterial cultures we use? ______

Laboratory Report 54

Student: ______________________

Desk No.: __________ Section: __________

O/F Gram-Negative Rods Identification: The Oxi/Ferm Tube II System

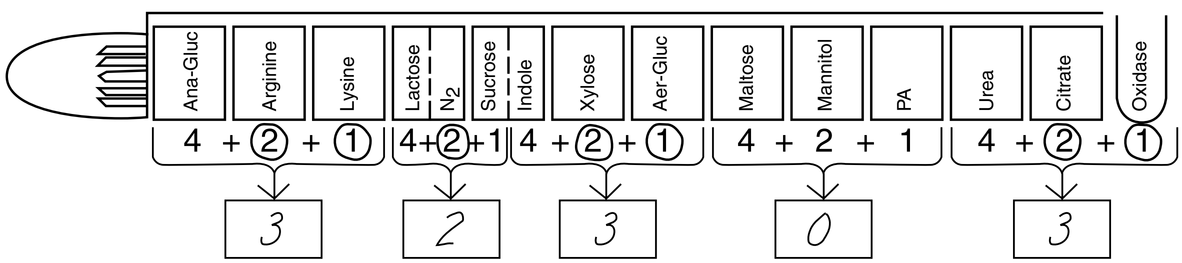
A. *Tabulation of Results and Code Determination*

Once you have marked the positive reactions on the side of the tube and circled the numbers that are assigned to each of the positive chambers, as indicated in the example below, add the numbers in each bracketed group to get the five-digit code.

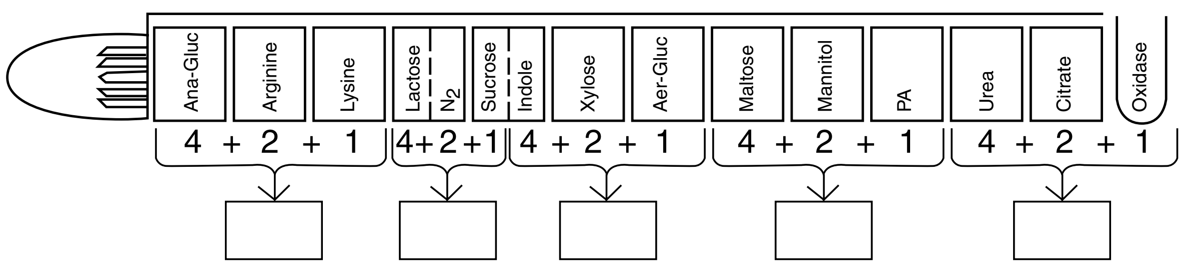
The final step is to look up the code number in the *Oxi/Ferm Tube II Biocode Manual* to determine the genus and species. If confirmatory tests are necessary, the manual will tell you which ones to perform.

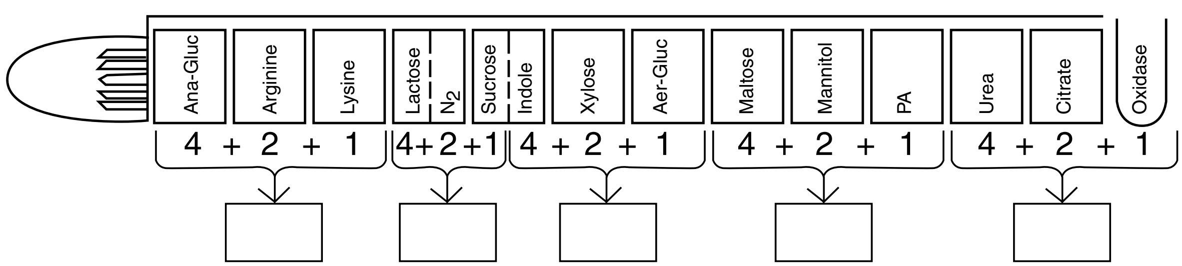
In the example below, the code number is 32303. If you look up this number in the *Biocode Manual* you will find on page 25 that the organism is *Pseudomonas aeruginosa.*

Use this procedure to identify your unknown by applying your results to the blank diagrams provided.



B. *Results Pads*





O/F Gram-Negative Rods Identification: The Oxi/Ferm Tube II System

C. Questions

1. What is the intended function of the Oxi/Ferm Tube II System? ______________________

2. In the "real world" who would use this system? ______________________

3. What might be an explanation for the failure of this system to work with some of the bacterial cultures we use? ______________________

Laboratory Report 55

Student: ______________________

Desk No.: ________ Section: ________

Staphylococcus Identification: The API Staph-Ident System

A. Tabulation of Results

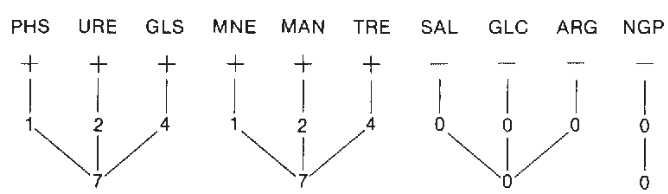
By referring to charts V and VI, Appendix D, determine the results of each test, and record these results as positive (+) or negative (−) in the Profile Determination Table below. Note that two more of these tables have been printed on the next page for tabulation of additional organisms.

	PHS 1	URE 2	GLS 4	MNE 1	MAN 2	TRE 4	SAL 1	GLC 2	ARG 4	NGP 1
RESULTS										
PROFILE NUMBER										

				Additional Information	Identification
GRAM STAIN		COAGULASE			
MORPHOLOGY		CATALASE			

B. Construction of Four-Digit Profile

Note in the above table that each test has a value of 1, 2, or 4. To compute the four-digit profile for your unknown, total up the positive values for each group.



C. Final Determination

Refer to the Staph-Ident Profile Register (chart VII, Appendix D) to find the organism that matches your profile number. Write the name of your unknown in the space below and list any additional tests that are needed for final confirmation. If the materials are available for these tests, perform them.

Name of Unknown: ______________________

Additional Tests: ______________________

Styphylococcus Identification: The API Staph-Ident System

	PHS 1	URE 2	GLS 4	MNE 1	MAN 2	TRE 4	SAL 1	GLC 2	ARG 4	NGP 1
RESULTS										
PROFILE NUMBER										

GRAM STAIN

MORPHOLOGY

COAGULASE

CATALASE

Additional Information

Identification

	PHS 1	URE 2	GLS 4	MNE 1	MAN 2	TRE 4	SAL 1	GLC 2	ARG 4	NGP 1
RESULTS										
PROFILE NUMBER										

GRAM STAIN

MORPHOLOGY

COAGULASE

CATALASE

Additional Information

Identification

D. *Questions*

1. What is the intended function of the API Staph-Ident System? ______________________________

2. In the "real world" who would use this system? ______________________________

3. What might be an explanation for the failure of this system to work with some of the bacterial cultures we use? ______________________________

LABORATORY REPORT 56, 57

Student: ______________________

Desk No.: __________ Section: __________

Ex. 56 Microbial Population Counts of Soil

A. *Tabulation of Results*

Select the best plate and count the organisms on a Quebec colony counter. Tabulate your results and the results of other students near you who cultured the other types of soil organisms. Be sure to count only representative types.

ORGANISMS	COUNT PER PLATE	DILUTION	ORGANISMS PER GRAM OF SOIL
Bacteria			
Actinomycetes			
Molds			

B. *Conclusions*

What generalizations can you make from this exercise?

____

____

____

____

C. *Questions*

1. Why would the number of bacteria present in the soil actually be higher than the number determined by your plate count? ______________________

____

____

2. What other types of microbes might be present in your soil sample? ______________________

____

____

Ex. 57 Isolation of an Antibiotic Producer from Soil

A. *Results*

Describe in detail how the experiment turned out.

B. *Questions*

1. What four genera of microbes produce the vast majority of antibiotics?

2. What is known about the importance of these antibiotics in the natural habitat of these microbes?

LABORATORY REPORT 59

Student: ____________________

Desk No.: __________ Section: __________

Nitrogen Fixing Bacteria

A. Questions

1. What enzyme is responsible for nitrogen fixation? ____________________
2. What metal is essential for this enzyme to function? ____________________
3. Why is nitrogen fixation so important? ____________________

____

B. Azotobacteraceae

Identify the characteristics of your isolate that led to your decision as to its identification.

1. Were the colonies pigmented? ____________________

   If pigmented, what color? ____________________

   Was fluorescence present? ____________________

   If fluorescent, what color? ____________________
2. Were you able to see cysts? ____________________
3. Is the organism motile? ____________________
4. Gram reaction? ____________________
5. Any other pertinent characteristic? ____________________

____

____

Name of Organism: ____________________

Draw a few cells of organisms in circle at right.

Azotobacteraceae

C. Rhizobiaceae

Draw a few representative cells of *Rhizobium* in the circle to the right.

1. What is the most important criterion for species identification in the family of nitrogen-fixers?

   ____

   ____
2. From the standpoint of amount of nitrogen fixation, is this group of nitrogen-fixers **more** or **less important** than the Azotobacteraceae?

   ____

Rhizobiaceae

LABORATORY REPORT 60

Student: ____________________

Desk No.: ________ Section: __________

Ammonification in Soil

A. Tabulation of Results

Record the presence or absence of ammonia in the tubes of media:

+	slight ammonia (faint yellow)
++	moderate ammonia (deep yellow)
+++	much ammonia (brown precipitate)
−	no ammonification

INCUBATION TIME	AMOUNT OF AMMONIA		pH	
	Control	Peptone with Soil	Control	Peptone with Soil
4 days				
7 days				

B. Questions

1. In the natural situation, what compounds serve as the source of the ammonia released in ammonification?

2. In simple terms, what occurs during decay?

3. Differentiate between the following:

   Peptonization:

   Ammonification:

4. What happens to ammonia in soil when **denitrification** of ammonia takes place?

LABORATORY REPORT

61, 62

Student: ______________________

Desk No.: __________ Section: __________

Ex. 61 Isolation of a Denitrifier from Soil
Using Nitrate Agar

A. Observations:

1. **Second Period.** Describe the types of colonies observed growing on the nitrate agar plate. ______

2. **Third Period.** What evidence do you have of the presence of a denitrifier in the soil sample you are working with? ______

3. **Microscopic Examination.** Describe the appearance of organisms that were observed on a gram-stained slide. ______

4. **Further Testing.** If you performed any further tests for species identification, state any conclusions that were made. ______

B. Questions:

1. Why are denitrifying bacteria costly to farmers? ______

2. Why are denitrifying bacteria essential to the existence of life on plant earth? ______

3. Of what value is denitrification to the organism? ______

Ex. 62 Isolation of a Denitrifier from Soil: Using an Enrichment Medium

A. *Observations:*

1. **First Enrichment Culture Appearance.** Describe the appearance of the first enrichment culture in the second laboratory period: ______

2. **Microscopic Examination:** Describe the appearance of the cells as seen on:

   Wet Mount: ______

   Gram-stained Slide: ______

3. **Second Enrichment Culture Appearance.** Describe the appearance of the second enrichment culture after five days incubation: ______

4. **Agar Plate Colonies.** Describe the characteristics of the colonies on the nitrate succinate-mineral salts agar: ______

   Do these characteristics match those of *Paracoccus denitrificans?* ______

5. **Final Gram-stained Slide.** How do the microscopic characteristics of your organism match the description of *Paracoccus denitrificans?* ______

B. *Questions:* Answer the questions that are provided on the front page of this Laboratory Report.

Laboratory Report 63

Student: ______________________

Desk No.: __________ Section: ____________

Bacteriological Examination of Water: Qualitative Tests

A. *Results of Presumptive Test (MPN Determination)*

Record the number of positive tubes on the chalkboard and on the following table. When all students have recorded their results with the various water samples, complete this tabulation. Determine the MPN according to the instructions on page 224.

WATER SAMPLE (SOURCE)	NUMBER OF POSITIVE TUBES				MPN
	3 Tubes DSLB 10 ml	3 Tubes SSLB 1.0 ml	3 Tubes SSLB 0.1 ml	3 Tubes SSLB 0.01 ml	

B. *Results of Confirmed Test*

Record the results of the confirmed tests for each water sample that was positive on the presumptive test.

WATER SAMPLE (SOURCE)	POSITIVE	NEGATIVE